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GEOGRAPHY · Major landforms and their features

UNDERSTAND IT FIRST

What is it? Landforms are the natural shapes of the Earth's surface: **mountains, plateaus, plains, valleys, dunes, caves** and so on. Two kinds of forces make them. **Endogenic (internal) forces** from inside the Earth **build up** land (folding, faulting, volcanoes). **Exogenic (external) forces** like **running water, wind, glaciers, waves and groundwater wear it down** and **deposit** material elsewhere. Exams ask you to **match landforms with the agent that makes them** and to give **Indian examples**. Learn it as **forces → weathering → mountains, plateaus, plains → landforms made by each agent → AP examples**.

1. Forces and weathering

- **Endogenic forces: diastrophism**, which includes **orogenic** movements (**mountain building** by folding and faulting) and **epeirogenic** movements (slow **uplift or sinking** of large areas), plus **volcanism** and **earthquakes**. **Exogenic forces: weathering**, **mass movement** (landslides) and **erosion and deposition** by agents; together they **level the land** (**denudation**, **gradation**).
- **Weathering** is the **breaking of rocks in place** (no movement). **Physical** weathering: heating and cooling make rock layers peel off (**exfoliation**, 'onion weathering', common in granite hills); water freezing in cracks (**frost wedging**). **Chemical** weathering: **oxidation** (rusting, red soils), **carbonation** (rainwater with CO_2 dissolves **limestone**), hydration. **Biological** weathering: roots and burrowing animals.

2. Mountains, plateaus and plains

Landform	Types and examples
Mountains	**Fold** mountains, formed when plates push rocks into folds: young ones like the **Himalayas, Alps, Andes, Rockies**; old ones worn down like the **Aravallis** and Appalachians. **Block (fault)** mountains: a block raised between faults is a **horst** (**Vosges, Black Forest**; the **Satpura** is often cited); a block that sinks forms a **rift valley (graben)** (**Narmada and Tapi valleys**, the Rhine valley, the Great Rift Valley). **Volcanic** mountains (**Fuji, Kilimanjaro**). **Residual** mountains, the worn remains of old ranges (**Aravallis, Eastern Ghats**). **Mount Everest** is **8,849 m** (2020 survey); the highest peak of the Eastern Ghats is **Arma Konda** (about **1,680 m**, AP); the highest in south India is **Anamudi** (**2,695 m**).
Plateaus	Flat-topped highlands. **Intermontane** (surrounded by mountains; **Tibet**, the highest, 'roof of the world'); **piedmont** (at the foot of mountains; **Patagonia**); **continental** (large raised areas; the **Deccan Plateau**, built of lava in its north-west). The **Chota Nagpur** plateau is India's **mineral heartland**. In AP, the **Rayalaseema** uplands are part of the Deccan Plateau.
Plains	Low, flat land, mostly made by **river deposits** (**alluvial plains** such as the **Indo-Gangetic plain**); **coastal plains** (the **east coast** is wider, with deltas); **peneplains** are almost flat surfaces left after long erosion. Plains support **dense population and farming**.

3. Landforms made by rivers

A river's life

- **Upper course (youth):** the river cuts **down** fast, making **V-shaped valleys**, deep **gorges** and **canyons** (the **Grand Canyon**), **waterfalls** (**Jog Falls** on the Sharavati), **rapids** and **potholes**. **Middle course (maturity):** the river cuts **sideways** and swings in loops called **meanders** across a **flood plain**. **Lower course (old age):** a meander is cut off to form an **oxbow lake**; floods build raised banks called **natural levees**; the river splits into **distributaries**.
- **Delta vs estuary:** a **delta** is a triangle of silt at the mouth, formed where a slow river drops its load into a calm sea (**Ganga–Brahmaputra**, the world's largest; **Krishna, Godavari, Mahanadi, Kaveri**). An **estuary** is a funnel-shaped mouth where tides wash the silt away (west-flowing **Narmada and Tapi**). **AP waterfalls: Talakona** (Seshachalam hills; the highest in AP), **Ethipothala** (on the Chandravanka, near Nagarjuna Sagar) and **Katiki** (Araku valley).

4. Landforms made by wind, glaciers, waves and groundwater

Other agents and their landforms

Agent	Erosional and depositional landforms
Wind (deserts)	Sand-laden wind cuts the base of rocks into **mushroom (pedestal) rocks**; isolated hills are **inselbergs**; ridges are **yardangs** and **zeugens**; **deflation hollows** form where sand is blown away. Deposits: **barchans** (**crescent-shaped** dunes), **seif** (long, **linear** dunes) and fine dust called **loess** (north China). The **Thar** has many shifting dunes.
Glaciers	Moving ice makes **U-shaped valleys**, armchair-shaped **cirques**, sharp **arêtes** and pyramid peaks called **horns** (the Matterhorn), **hanging valleys** and, where the sea floods a glacial valley, **fjords** (Norway). Deposits: **moraines** (heaps of rock debris), **eskers** (winding ridges), **drumlins** and **outwash plains**.
Sea waves	Erosion: **cliffs**, **sea caves**, **arches** and **stacks**. Deposition: **beaches**, **spits** and **bars**; a bar can enclose a **lagoon** (**Chilika**, Odisha, India's largest brackish lagoon; **Pulicat**, AP–TN, the second largest; **Vembanad**, Kerala). A **tombolo** joins an island to the land. AP's coast (about **974 km**, second longest after Gujarat) has beaches such as **RK Beach**, **Suryalanka** and **Mypadu**.
Groundwater (karst)	In **limestone**, water dissolves rock to form **sinkholes**, underground **caves**, **stalactites** (hanging from the **ceiling**: remember 'c') and **stalagmites** (rising from the **ground**: 'g'); when they meet they form **pillars**. Examples: **Borra Caves** (Ananthagiri hills, Alluri Sitharama Raju district; found by **William King, 1807**) and **Belum Caves** (Nandyal district; among the longest in India); India's longest cave is **Krem Liat Prah** (Meghalaya).

EXAM TRAP

Endogenic (orogenic, epeirogenic, volcanism) vs **exogenic** (weathering, erosion, deposition); **exfoliation** (granite), **frost wedging**, **carbonation** (limestone), **oxidation**; **fold** (Himalaya young; Aravalli old), **horst**

(Vosges, Black Forest, Satpura), **graben** (Narmada, Tapi, Rhine), **volcanic** (Fuji, Kilimanjaro), **residual** (Aravalli, Eastern Ghats); **Everest 8,849 m**; **Arma Konda ~1,680 m**; **Anamudi 2,695 m**; plateaus: **intermontane (Tibet)**, **piedmont (Patagonia)**, **continental (Deccan)**; Chota Nagpur minerals; river: **V-valley, gorge, waterfall, potholes → meander, flood plain → oxbow lake, levee, delta**; **estuary: Narmada, Tapi**; AP falls: **Talakona, Ethipothala, Katiki**; wind: **mushroom rock, inselberg, barchan (crescent), seif, loess**; glacier: **U-valley, cirque, horn, fjord, moraine, esker**; waves: **cliff, arch, stack, spit, bar, lagoon (Chilika, Pulicat)**; AP coast **~974 km**; karst: **sinkhole, stalactite (ceiling), stalagmite (ground)**; **Borra (1807), Belum**; **Krem Liat Prah**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Stalactites hang from the ceilings of limestone caves.

2. Oxbow lakes are formed in the lower (old) course of a river.

3. Barchans are crescent-shaped sand dunes.

4. Fjords are formed mainly by wind erosion.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Fjords are glacial valleys drowned by the sea.

PRACTICE

Match List-I (type of landform) with List-II (example):

List-I (type of landform)	List-II (example)
A. Young fold mountains	1. Aravallis
B. Horst (block mountain)	2. Narmada valley
C. Rift valley (graben)	3. Vosges
D. Residual mountains	4. Himalayas

(1) A → 3 | B → 4 | C → 2 | D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) Himalayas: young fold; Vosges: horst; Narmada: rift valley; Aravallis: residual.

PRACTICE

Which one of the following statements is NOT correct?

A. The Borra Caves are limestone caves in the Ananthagiri hills.

B. The Narmada forms a large delta where it enters the sea.

C. Chilika is India's largest brackish-water lagoon.

D. The Tibetan plateau is an intermontane plateau.

Answer: B. The Narmada flows through a rift valley and forms an estuary, not a delta.

PRACTICE

Assertion (A): Mushroom rocks are common in deserts.
Reason (R): In deserts, chemical weathering by heavy rainfall is the main force shaping rocks.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: deserts get little rain; sand blown by wind cuts the lower part of rocks, making mushroom shapes.

PRACTICE

A trekker in a high mountain area sees a U-shaped valley, a bowl-shaped hollow holding a small lake, and long ridges of loose rock debris. Which agent most likely made these landforms?

A. Running water

B. Wind

C. Glacier

D. Sea waves

Answer: C. U-shaped valleys, cirques (with tarn lakes) and moraines are glacial landforms.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write endogenic and exogenic forces and the three types of weathering. 2. Write mountains, plateaus and plains with types and examples. 3. Write landforms by agent (river stages, wind, glacier, waves, karst) with AP examples.

ECONOMY · National income: concept and measurement

UNDERSTAND IT FIRST

What is it? National income is the **total value of all goods and services** a country produces in a year, counted in money. It tells us **how big the economy is, how fast it is growing** and, divided by population, **how rich people are on average (per capita income)**. Governments use it to plan budgets; news reports on **'GDP growth'** every quarter. The terms (GDP, GNP, NNP, factor cost) confuse many students, but they follow a simple chain of **add or subtract one item**. Learn it as **basic terms → the GDP-to-NI chain → prices (market, factor, basic; nominal, real) → three methods → what is counted → India's history → limits and alternatives**.

1. Basic terms and the chain

From GDP to National Income

Term	Meaning
GDP (Gross Domestic Product)	Market value of all **final** goods and services produced **within the country's borders** in a year, by citizens or foreigners. **Final goods** are bought for final use; **intermediate goods** (raw materials used up in production) are **not counted separately** to avoid **double counting**.
GNP (Gross National Product)	**GDP + net factor income from abroad (NFIA)**. NFIA = what **Indians earn abroad** (wages, interest, profits) minus what **foreigners earn in India**. GNP is about **nationals**, GDP about **territory**. For India NFIA is usually **negative**, so GNP is slightly **less** than GDP.
NDP and NNP	**Net = Gross − depreciation** (consumption of fixed capital, the wear and tear of machines and buildings). **NDP = GDP − depreciation; NNP = GNP − depreciation.**

Market price vs factor cost	**Factor cost** is what producers actually receive for land, labour, capital and enterprise. **Factor cost = market price − indirect taxes + subsidies** (that is, minus **net indirect taxes**). **National Income = NNP at factor cost.** Since **2015**, India reports **GVA at basic prices** (which includes production taxes but not product taxes like GST); **GDP = GVA at basic prices + product taxes − product subsidies**.
Nominal vs real; per capita	**Nominal GDP** uses **current prices**; **real GDP** uses **constant (base-year) prices**, so it shows real growth without inflation. **GDP deflator = (nominal GDP ÷ real GDP) × 100**, a broad measure of inflation. **Per capita income = national income ÷ population. Personal income** adds transfer payments; **disposable income** = personal income − direct taxes.

2. Three methods of measurement

Method	How it works
Product (value-added) method	Adds the **value added** at each stage of production in every sector (output minus intermediate inputs). Example: a farmer sells wheat for ₹50, the miller sells flour for ₹70, the baker sells bread for ₹100: value added is 50 + 20 + 30 = **₹100**, equal to the final good's price.
Income method	Adds incomes earned by factors: **compensation of employees (wages) + rent + interest + profit + mixed income** of the self-employed.
Expenditure method	Adds spending on final goods: **GDP = C + I + G + (X − M)**: private **consumption**, **investment**, **government** spending and **net exports**.

- **Included: imputed rent** of owner-occupied houses, **own-account production** (farm produce a farmer keeps), the **commission** on sale of second-hand goods. **Excluded: transfer payments** (old-age pensions, scholarships, unemployment allowance), **sale of second-hand goods** (already counted when new), **purely financial transactions** (buying shares), **illegal activities**, and **unpaid household work** by family members.
- **India's history: Dadabhai Naoroji** made the first estimate (**1867–68**, about **₹20 per head**) in **'Poverty and Un-British Rule in India'**; later William Digby and Findlay Shirras; **V.K.R.V. Rao** made the first **scientific** estimate (**1931–32**). The **National Income Committee (1949)** was chaired by **P.C. Mahalanobis** (members **D.R. Gadgil, V.K.R.V. Rao**). The **Central Statistical Organisation (CSO, 1951)** prepared estimates; in **2019** the CSO and NSSO merged into the **National Statistical Office (NSO)** under **MoSPI**. India uses mainly the **production and income methods**. The base year has been **2011–12**; MoSPI is moving the series to a new base year, **2022–23** (check the latest release).
- **Limits of GDP and alternatives:** GDP does not show **inequality, environmental damage, leisure, unpaid work or the informal sector** well, and it counts harmful spending as growth. Alternatives: **Human Development Index (UNDP, 1990; Mahbub ul Haq, with Amartya Sen)**: **health (life expectancy), education (schooling) and income (GNI per capita, PPP)**; India ranked about **130th of 193** in the 2025 report; **Green GDP** (subtracting environmental loss); **Gross National Happiness (Bhutan)**; the **Multidimensional Poverty Index**. **AP:** the State's **Socio-Economic Survey** reports **GSDP**, sector shares and **per capita income** every year.

EXAM TRAP

GDP = final goods within borders; **GNP = GDP + NFIA** (India's NFIA usually negative); **NDP/NNP = gross − depreciation**; **FC = MP − indirect taxes + subsidies**; **NI = NNP at FC**; **GVA at basic prices (2015)**; GDP = GVA + product taxes − product subsidies; **nominal (current) vs real (constant)**; **deflator = nominal ÷ real × 100**; per capita = NI ÷ population; methods: **product (value added), income (wages + rent + interest + profit + mixed), expenditure (C + I + G + X − M)**; include **imputed rent, own-account output, commission**; exclude **transfers, second-hand goods, shares, illegal, household work**; **Naoroji 1867–68 (₹20)**; **V.K.R.V. Rao 1931–32**; **NIC 1949 (Mahalanobis)**; **CSO 1951 → NSO 2019 (MoSPI)**; base **2011–12 → 2022–23**; **HDI 1990 (Mahbub ul Haq)**; India **~130/193 (2025)**; Green GDP, GNH, MPI.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. GNP = GDP + net factor income from abroad.

2. NNP = GNP − depreciation.

3. National income is NNP at factor cost.

4. Old-age pensions paid by the government are included in national income.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. Pensions are transfer payments: no goods or services are produced in return, so they are excluded.

PRACTICE

Match List-I (person) with List-II (contribution):

List-I (person)	List-II (contribution)
A. Dadabhai Naoroji	1. Human Development Index
B. V.K.R.V. Rao	2. Chairman, National Income Committee, 1949
C. P.C. Mahalanobis	3. First scientific estimate, 1931–32
D. Mahbub ul Haq	4. First estimate of national income, 1867–68

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Naoroji: first estimate; Rao: first scientific; Mahalanobis: 1949 Committee; Mahbub ul Haq: HDI.

PRACTICE

Which one of the following statements is NOT correct?

A. Real GDP is measured at constant (base-year) prices.

B. GDP deflator = (nominal GDP ÷ real GDP) × 100.

C. The imputed rent of owner-occupied houses is included in GDP.

D. The full sale value of a second-hand car is included in this year's GDP.

Answer: D. A used car was counted when it was new; only the dealer's commission is new value added.

PRACTICE

Assertion (A): A country's GDP can rise while the welfare of many of its people does not improve.
Reason (R): GDP is usually calculated at market prices.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but R does not explain A. Welfare may not improve because GDP ignores inequality, environmental damage and unpaid work.

PRACTICE

A country's GDP at market prices is ₹1,000 crore. Depreciation is ₹100 crore, net factor income from abroad is −₹20 crore and net indirect taxes are ₹80 crore. What is its national income (NNP at factor cost)?

A. ₹780 crore

B. ₹800 crore

C. ₹880 crore

D. ₹820 crore

Answer: B. 1,000 − 20 (NFIA) = 980 GNP; 980 − 100 = 880 NNP at MP; 880 − 80 = 800.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the GDP → GNP → NNP → NI chain with what is added or subtracted. 2. Write the price concepts and the three methods with formulas. 3. Write what is included and excluded, India's history and base year, and the limits and alternatives.

SCIENCE · Wildlife conservation: IUCN, CITES, endangered species, biosphere reserves

UNDERSTAND IT FIRST

What is it? Yesterday we saw why biodiversity is being lost. Today is **how the world and India protect wild animals and plants**: by **grading how endangered each species is** (IUCN Red List), **controlling international trade** in wildlife (CITES), **strict laws** (Wildlife Protection Act) and **protected areas** such as national parks, tiger reserves and **biosphere reserves**. Learn it as **IUCN categories → CITES and other conventions → Indian laws → protected areas and projects → biosphere reserves → AP's protected areas**.

1. The IUCN Red List

IUCN Red List categories

- The **International Union for Conservation of Nature (IUCN)** was founded in **1948** (headquarters **Gland, Switzerland**); its **Red List** (since **1964**) is the world's main record of species' risk. **'Threatened'** means **Critically Endangered, Endangered or Vulnerable**. It is a **scientific list, not a law**.
- **Indian examples: Critically Endangered**: **Great Indian Bustard**, **gharial**, **Jerdon's courser**, **white-rumped vulture**; **Endangered**: **Bengal tiger**, **Asian elephant**, **Ganges river dolphin**, **red panda**; **Vulnerable**: **snow leopard**, **one-horned rhinoceros**. The **Great Indian Bustard** is threatened mainly by **collisions with overhead power lines** in Rajasthan and Gujarat.

2. CITES and other conventions

Convention	Facts
CITES (Washington Convention)	Convention on International Trade in Endangered Species of Wild Fauna and Flora; signed **3 March 1973** (now **World Wildlife Day**), in force **1975**; **India joined in 1976**; secretariat in **Geneva** (under UNEP). It is **legally binding** on members but works **through national laws**. **Appendix I**: species threatened with extinction; **commercial trade banned** (tiger, Asiatic lion, most rhinos). **Appendix II**: trade allowed **only with permits** (e.g. **red sanders**, many corals). **Appendix III**: species protected in **one country** that asks others to help control trade. **CoP20** was held at **Samarkand, Uzbekistan (2025)**.
CMS (Bonn Convention)	Conservation of **Migratory Species** (**1979**); India hosted **CoP13** at **Gandhinagar (2020)**, mascot 'Gibi' the Great Indian Bustard.
Ramsar Convention	For **wetlands** (Ramsar, **Iran, 1971**); India joined in **1982** (first sites **Chilika** and **Keoladeo**) and has **over 90** Ramsar sites; AP's site is **Kolleru**. **2 February** is World Wetlands Day.
Other bodies	**WWF (1961)**; **TRAFFIC** (monitors wildlife trade); **International Whaling Commission (1946)**; **Global Tiger Forum**; the **International Big Cat Alliance (2023)**, launched by India.

3. Indian laws, protected areas and projects

- **Constitution:** the **42nd Amendment (1976)** moved **forests** and **protection of wild animals and birds** from the State List to the **Concurrent List**, and added **Art 48A** (State to protect environment and wildlife) and **Art 51A(g)** (citizens' duty to protect nature and have compassion for living creatures).
- **Wildlife (Protection) Act, 1972:** bans hunting of listed animals; set up **National Parks** and **Wildlife Sanctuaries**; the **2002 amendment (in force 2003)** added **Conservation Reserves** and **Community Reserves**; the **2006 amendment** created the **National Tiger Conservation Authority (NTCA)** and the **Wildlife Crime Control Bureau (WCCB)**; the **2022 amendment** cut the schedules from six to **four** (**Schedule I** highest protection, **II** lesser, **III** plants, **IV CITES** specimens) and gave effect to CITES. The **National Board for Wildlife** is chaired by the **Prime Minister**. A **national park** allows **no human activity** such as grazing; a **sanctuary** may allow some.
- **Projects: Project Tiger (1 April 1973**, started with **9 reserves**; now about **58**; the **2022 census** estimated about **3,682 tigers**, around **three-quarters of the world's wild tigers**); **Project Elephant (1992)**; **Project Dolphin (2020)** (the **Gangetic dolphin** is the **national aquatic animal**); **Project Cheetah (2022, Kuno)**; **vulture** conservation (veterinary **diclofenac banned, 2006**). India's **largest tiger reserve** is **Nagarjunasagar–Srisailam** in AP.

4. Biosphere reserves

Zones of a biosphere reserve

- **Biosphere reserves** come from **UNESCO's Man and the Biosphere (MAB) Programme (1971)**. They protect **whole ecosystems together with the people who live in them**, using three zones: a strictly protected **core**, a **buffer** for research and education, and a **transition** zone with villages and sustainable use.

- **India** has **18 biosphere reserves**, declared by the Union Government. The first was **Nilgiri (1986)**; the largest is the **Great Rann of Kutch**. **13** are in **UNESCO's World Network**: **Panna** joined in **2020**, and **Cold Desert** (Himachal Pradesh) in **2025** (check the latest). **AP** has the **Seshachalam Biosphere Reserve (2010)**, home of **red sanders**, which is **not yet** in the UNESCO network.

AP's protected areas (all regions): Nagarjunasagar–Srisailam Tiger Reserve (Nallamala); **Papikonda National Park** (Godavari gorge); **Sri Venkateswara National Park** (Seshachalam); **Coringa** and **Krishna** wildlife sanctuaries (mangroves); **Kolleru** (wetland birds); **Pulicat (Nelapattu)** bird sanctuary (pelicans, flamingos); **Rollapadu** (Great Indian Bustard); **Sri Lankamalleswara** (Jerdon's courser); **Kambalakonda** (Visakhapatnam); **Telineelapuram and Telukunchi** (Srikakulam; pelicans and painted storks).

EXAM TRAP

IUCN 1948 (Gland); **Red List 1964**; threatened = **CR, EN, VU**; not a law; **CR: GIB, gharial, Jerdon's courser, white-rumped vulture**; **EN: tiger, Asian elephant, Ganges dolphin, red panda**; **VU: snow leopard, one-horned rhino**; **CITES 3 Mar 1973 / 1975**, **India 1976**, **Geneva**; **App I** (trade banned), **II** (permits; red sanders), **III** (one country); **CoP20 Samarkand 2025**; **CMS 1979 (CoP13 Gandhinagar 2020)**; **Ramsar 1971 / India 1982**, **90+** sites, **Kolleru**; **42nd Amendment 1976** (Concurrent List, **48A, 51A(g)**); **WPA 1972**; conservation and community reserves (**2003**); **NTCA, WCCB (2006 amendment)**; **2022: four schedules**; NBWL (PM); **Project Tiger 1973 (9 → ~58)**; **3,682 tigers (2022)**; **Elephant 1992**; **Dolphin 2020**; **Cheetah 2022**; **diclofenac ban 2006**; **NSTR largest**; **MAB 1971**; core, buffer, transition; **18 BRs**; **Nilgiri 1986** first; **Kutch** largest; **13 in UNESCO network** (Panna 2020, Cold Desert 2025); **Seshachalam BR 2010**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about CITES:
1. It was signed at Washington in 1973.
2. Commercial trade in Appendix I species is generally banned.
3. India is a party to CITES.
4. Its decisions automatically become law in member countries without any national legislation.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. CITES is binding but must be carried out through national laws; India did this through the Wildlife (Protection) Amendment Act, 2022.

PRACTICE

Match List-I with List-II:

List-I	List-II
A. Core zone	1. UNESCO, 1971
B. Buffer zone	2. Villages and sustainable use
C. Transition zone	3. Research and education allowed
D. Man and the Biosphere Programme	4. Strictly protected; no human activity

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Core: strict; buffer: research; transition: settlements; MAB: UNESCO 1971.

PRACTICE

Which one of the following pairs is NOT correctly matched?

A. Ramsar Convention : conservation of migratory species

B. Nagarjunasagar–Srisailam : India's largest tiger reserve

C. Nilgiri : India's first biosphere reserve

D. Project Tiger : 1973

Answer: A. Ramsar is for wetlands; migratory species come under the Bonn Convention (CMS).

PRACTICE

Assertion (A): The Great Indian Bustard is listed as Vulnerable on the IUCN Red List.

Reason (R): Collisions with overhead power lines are a major threat to the Great Indian Bustard.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: the Great Indian Bustard is Critically Endangered. R is true.

PRACTICE

Customs officers seize a parcel of pangolin scales being smuggled abroad from an airport. Which national body specialises in fighting such organised wildlife crime?

A. National Tiger Conservation Authority

B. Wildlife Crime Control Bureau

C. National Biodiversity Authority

D. Central Pollution Control Board

Answer: B. The WCCB, set up under the 2006 amendment, collects intelligence and coordinates action against wildlife trafficking.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the IUCN categories with Indian examples. 2. Write CITES with its three appendices and the other conventions. 3. Write the Constitution and the Wildlife Act points, the projects, biosphere reserve zones and India's figures, and AP's protected areas.

NOTEBOOK PAGE · Day 27: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Landforms

Area	Write
Forces and big landforms	Endogenic vs exogenic; weathering types; mountains (4 types with examples), plateaus (3), plains
Agents	River stages; delta vs estuary; wind, glacier, wave and karst landforms; AP examples from every region

ECONOMY: National income

- Definitions (GDP, GNP, NDP, NNP, NI, per capita); MP vs FC vs basic prices; nominal vs real, deflator; three methods; what is included and excluded; history (Naoroji to NSO), base year; limitations and alternatives

(HDI, Green GDP).

SCIENCE: Wildlife conservation

Write: IUCN categories with Indian examples; CITES appendices; other conventions; Wildlife Protection Act and its 2022 schedules; protected area types; projects with years; biosphere reserve zones and India's facts; AP's protected areas.